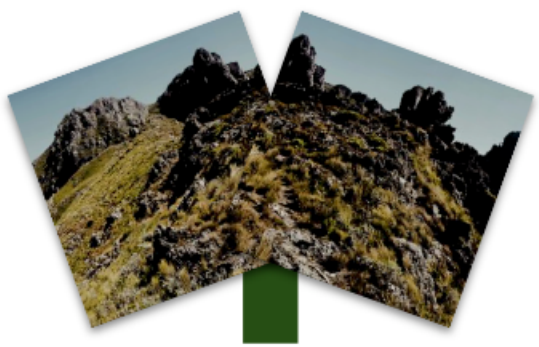
Large-scale training and multimodal vision

Research highlights

Nithin Gopalakrishnan Nair



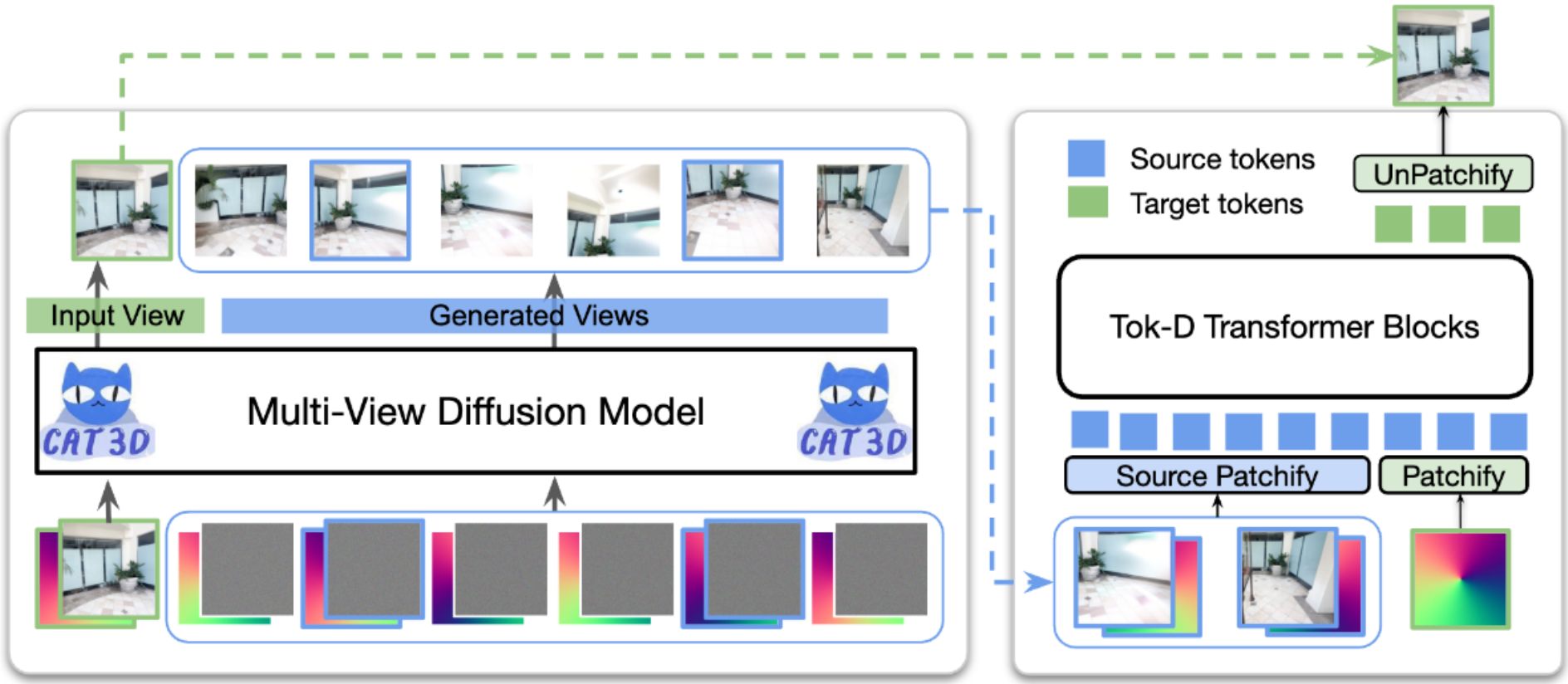




Ph.D., Johns Hopkins University

Google: training novel view synthesis

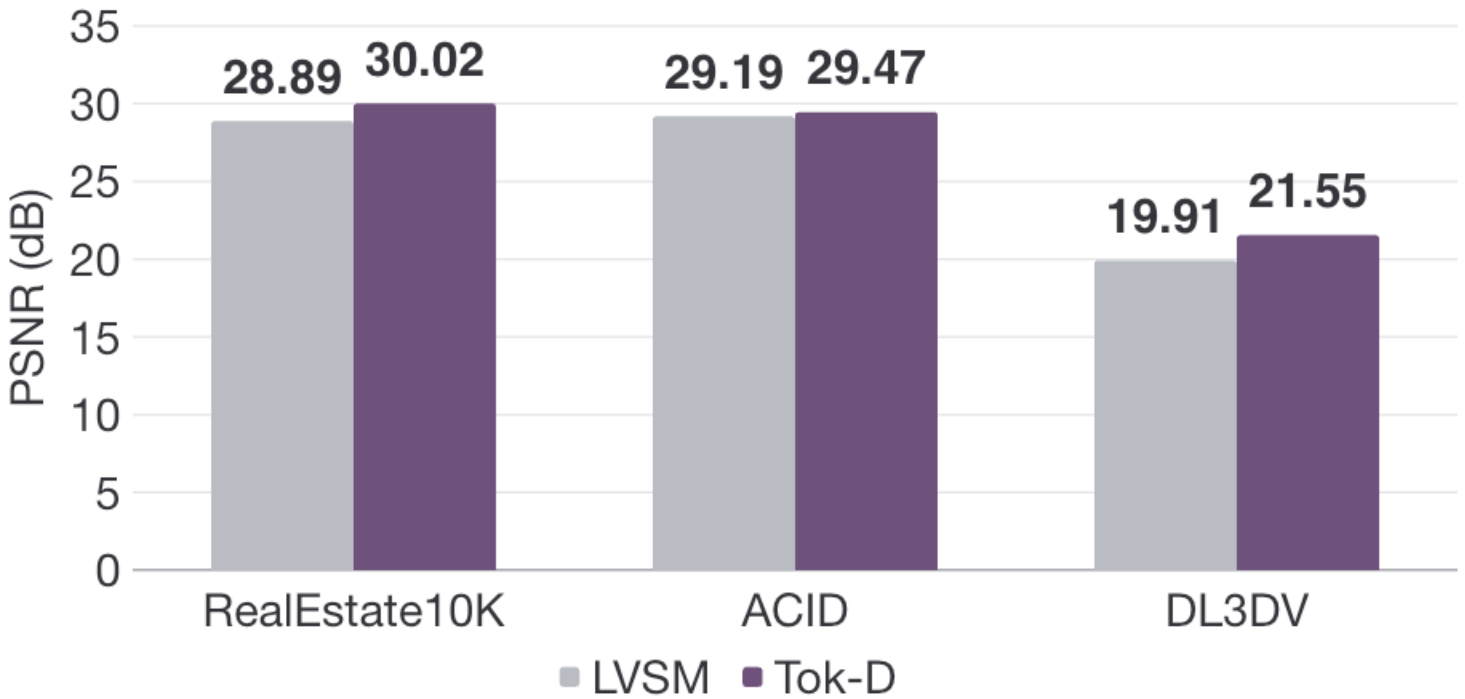
Diffusion-generated views expand the training data for transformer models.



ICCV 2025, co-first author. Training on 8 H100 GPUs.

Google: stronger reconstruction quality

In-distribution evaluation at 256 × 256 resolution. Higher PSNR is better.



+1.13 dB

RealEstate10K

+1.64 dB

DL3DV

ICCV 2025, Table 1. Real-data training comparison.

GenDeg: synthetic data for restoration

Diffusion models generate realistic degradations with controllable severity.

550K+

synthetic image pairs

Six degradation types
750K+ samples in GenDS

Better restoration on unseen degradation patterns.

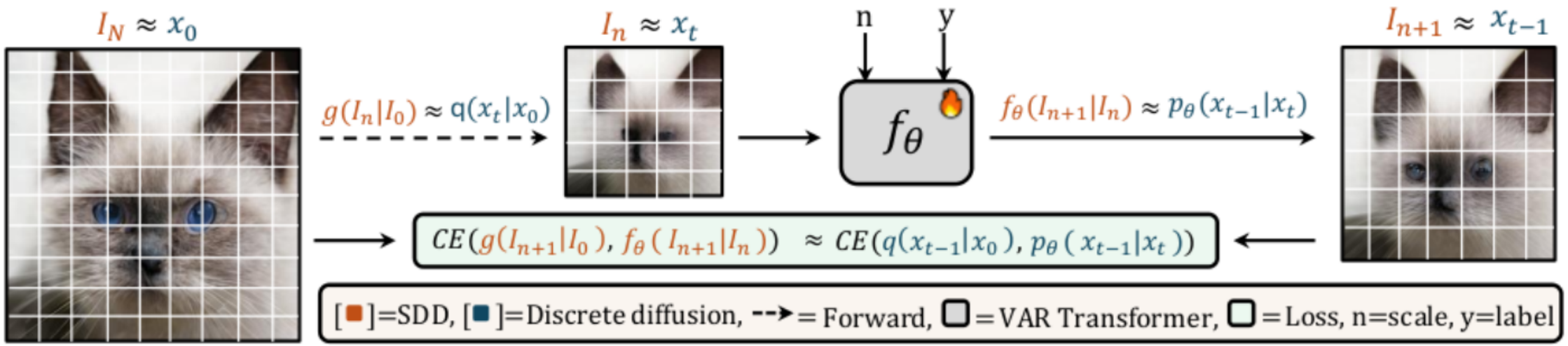


GD denotes training with GenDS.

GenDeg, CVPR 2025

Scale-Wise VAR is Secretly Discrete Diffusion

A connection between autoregressive visual generation and discrete diffusion.



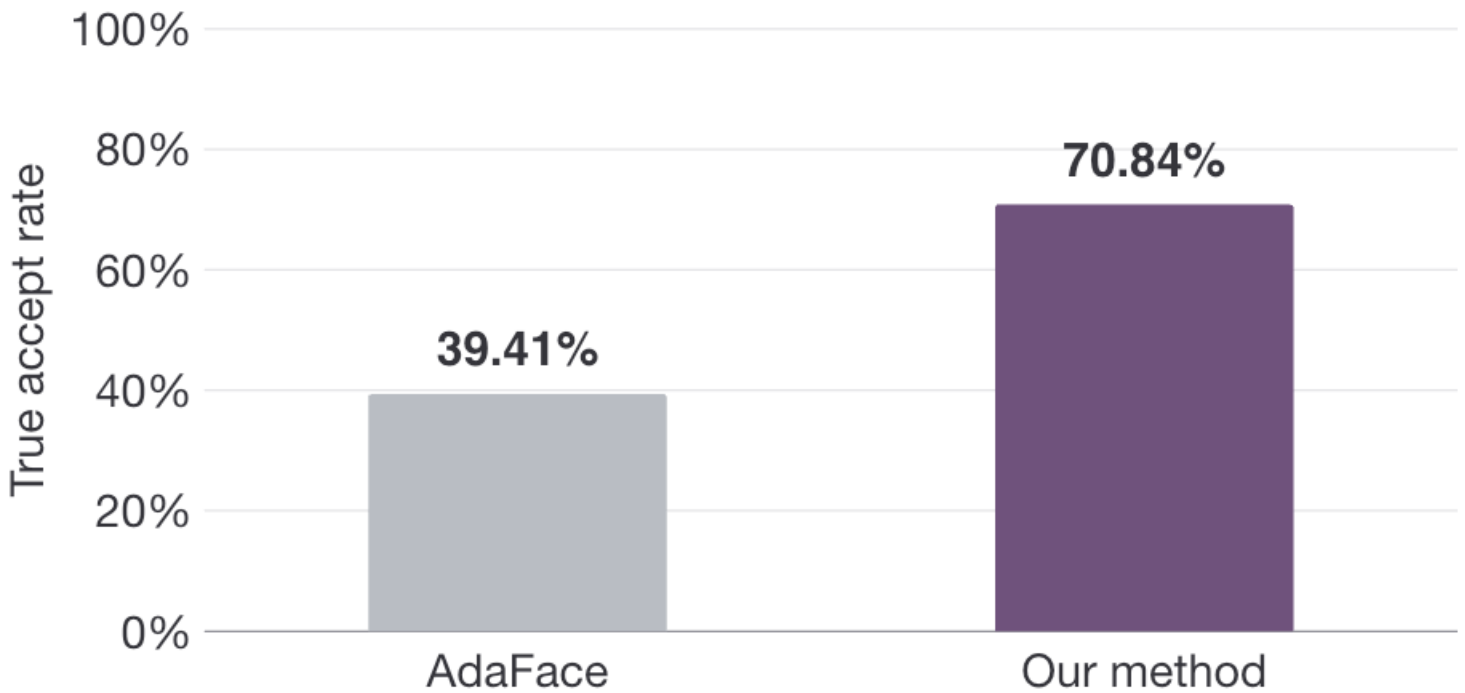
A Markovian attention mask establishes the equivalence.

SRDD uses this connection for iterative visual refinement.

arXiv preprint, 2025, co-first author

BRIAR: unconstrained face recognition

BRIAR verification at FAR 0.1%, with WebFace12M pretraining.



+31.4

percentage points

81.38%

Rank-1 with WebFace42M

FG 2025. AdaFace (ResNet101) versus our method (Swin-B).

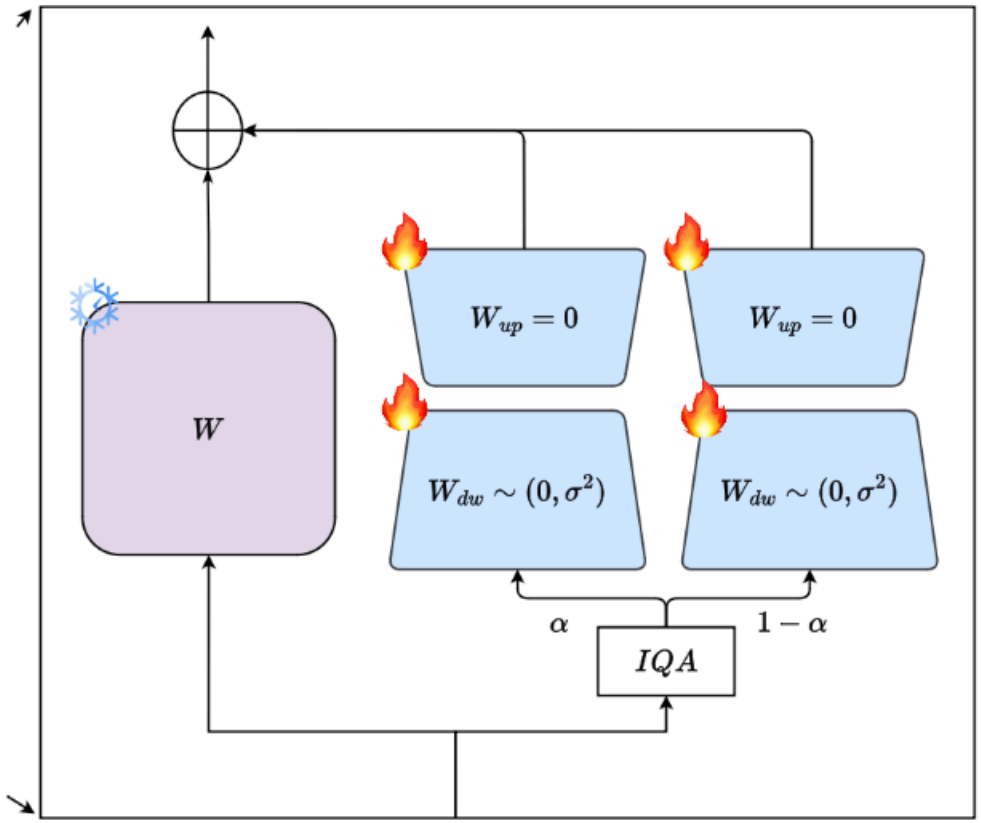
PETALface: adapting to low-resolution faces

Quality-weighted adapters preserve performance across input resolutions.

0.48%

trainable parameters

Two LoRA adapters adapt a frozen backbone to image quality.

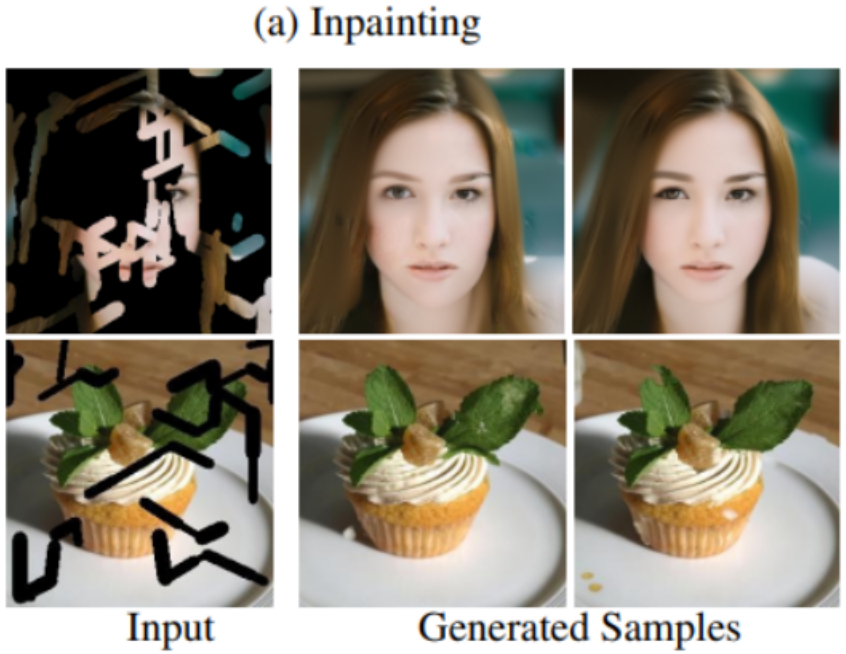


PETALface, WACV 2025 Oral

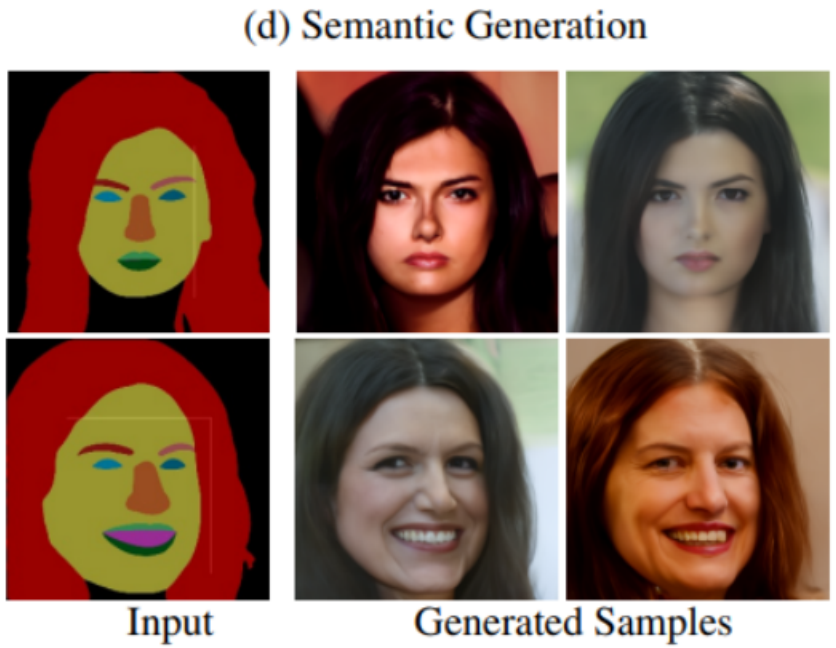
Steered Diffusion: inference-time control

First plug-and-play framework for image-to-image and label-conditioned synthesis.

Image-to-image generation



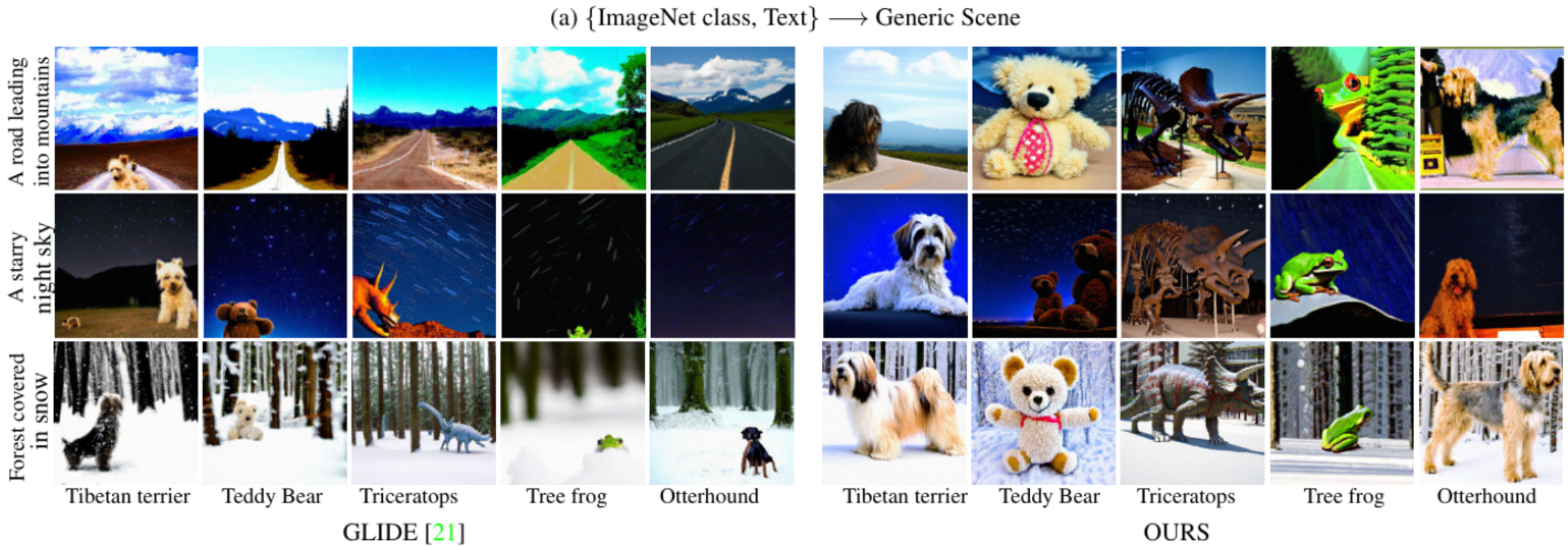
Label-conditioned generation



ICCV 2023, MERL and Johns Hopkins University

Unite and Conquer: compositional generation

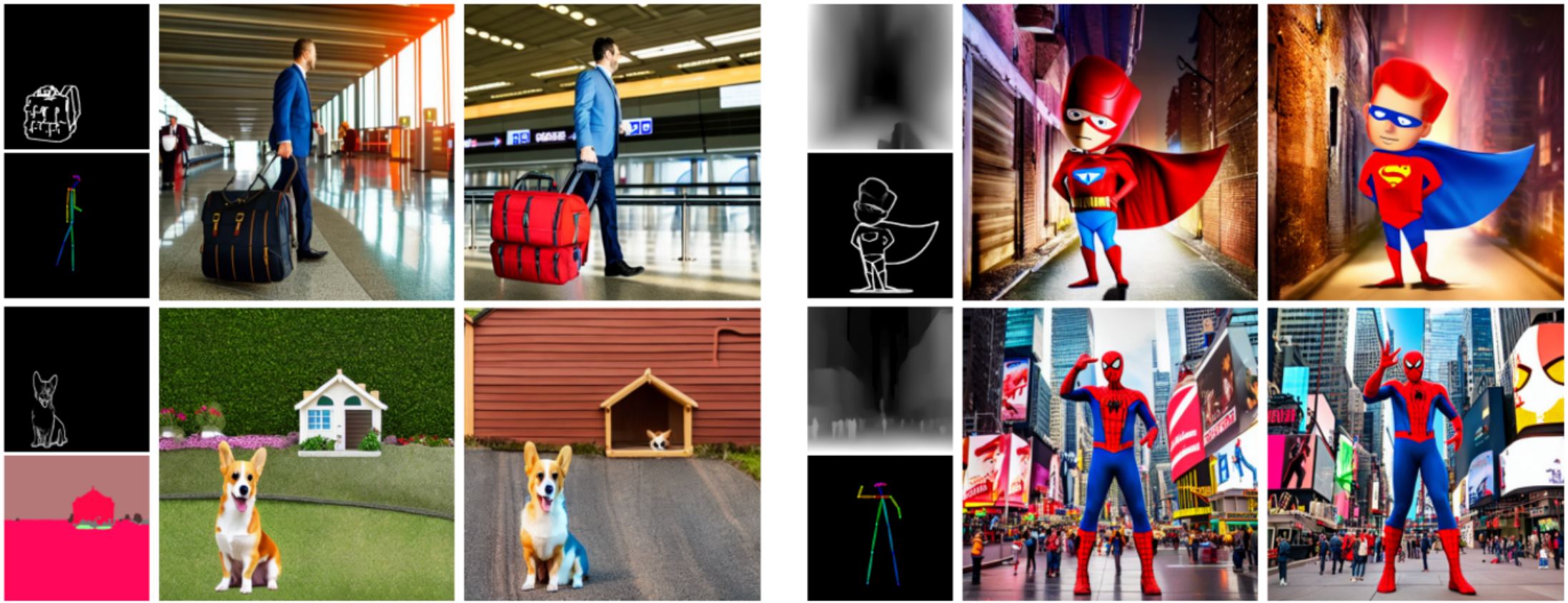
Pioneered multimodal generation by composing independent diffusion models.



CVPR 2023

MaxFusion: multiple spatial conditions

Variance-guided feature fusion combines off-the-shelf conditional models.



ECCV 2024

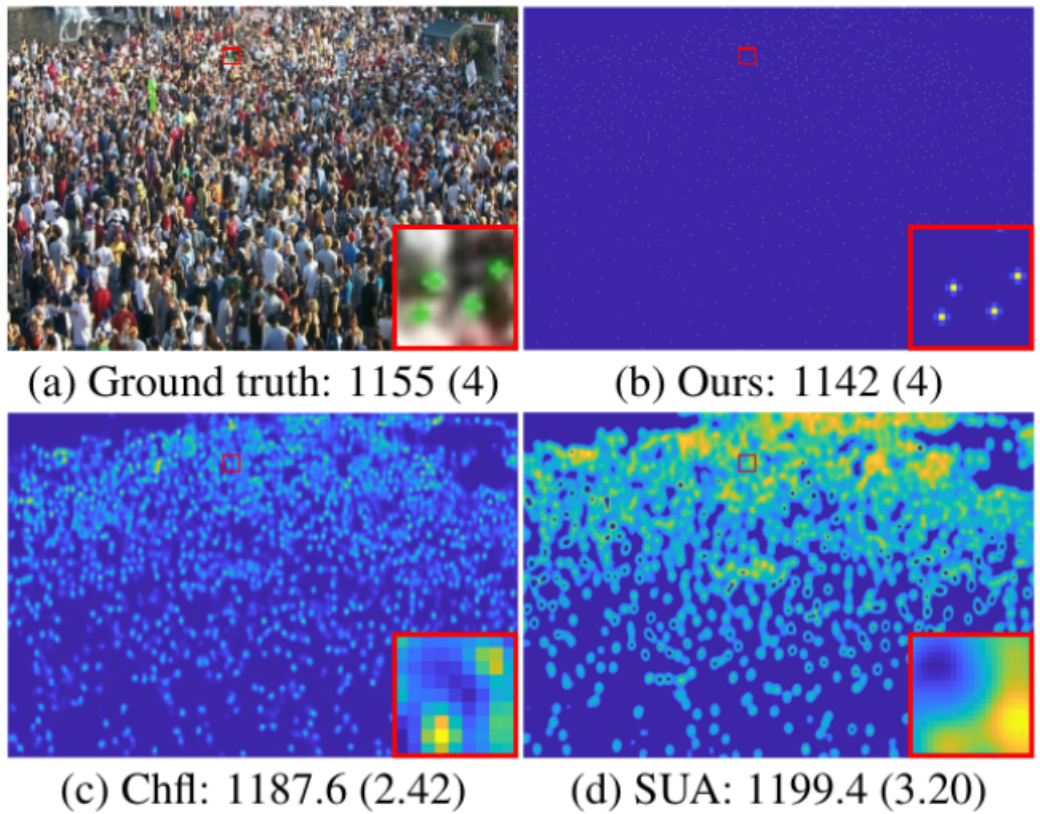
CrowdDiff: diffusion for crowd counting

Multiple density-map hypotheses improve crowd density estimation.

Conditional diffusion predicts detailed crowd density maps.

Multiple sampled hypotheses reduce ambiguity in counting.

One paper example shown at right.



(a) Ground truth: 1155 (4) (b) Ours: 1142 (4)

(c) Chfl: 1187.6 (2.42) (d) SUA: 1199.4 (3.20)

CrowdDiff, CVPR 2024

Research direction

Large-scale training of generative and multimodal models

Synthetic data expands what models can learn.
Inference-time control makes generation more useful.

18 peer-reviewed publications, including CVPR, ICCV and ECCV

nithin-gk.github.io